

Math 520, F15

Quiz 3

Name: Answer

SSN: xxx-xx-

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer.

Problem 1. (5 points) Given the following autonomous differential equation:

$$\frac{dp}{dt} = 8p - 2p^2$$

Find all critical points of the above equation and determine whether it is asymptotically stable or unstable for each of them, then find $\lim_{t \rightarrow +\infty} p(t)$ when $p(0) = p_0 > 0$.

Solution: $\frac{dp}{dt} = 8(p - \frac{p^2}{4}) = 8p(1 - \frac{p}{4})$

$$8p(1 - \frac{p}{4}) = 0 \Rightarrow p_1 = 0, p_2 = 4.$$

$\frac{dp}{dt} < 0$ $\frac{dp}{dt} > 0$ $\frac{dp}{dt} < 0$ Thus
 $\leftarrow$ $\rightarrow$ $\leftarrow$ $p_1 = 0 \rightarrow$ unstable
 $p_2 = 4 \rightarrow$ stable.

If $p(0) = p_0 > 0$, then

$$\lim_{t \rightarrow +\infty} p(t) = 4.$$

Problem 2. (5 points) Verify whether the given differential equation is exact and then solve it if yes: $(4x - y) dx + (6y - x) dy = 0$.

Solution: $M(x, y) = 4x - y \Rightarrow \frac{\partial M}{\partial y}(x, y) = -1 \Rightarrow \text{Exact!}$
 $N(x, y) = 6y - x \Rightarrow \frac{\partial N}{\partial x}(x, y) = -1$

Find ψ s.t. $\psi_x(x, y) = M(x, y)$ and $\psi_y(x, y) = N(x, y)$.

We have

$$\begin{aligned} \psi(x, y) &= \int M(x, y) dx = \int (4x - y) dx \\ &= 2x^2 - xy + h(y) \end{aligned}$$

$$\Rightarrow \psi_y(x, y) = -x + h'(y) = 6y - x$$

$$\Rightarrow h'(y) = 6y \Rightarrow h(y) = 3y^2$$

Thus $\psi(x, y) = 2x^2 - xy + 3y^2$ and the solution is implicitly given by

$$2x^2 - xy + 3y^2 = C$$